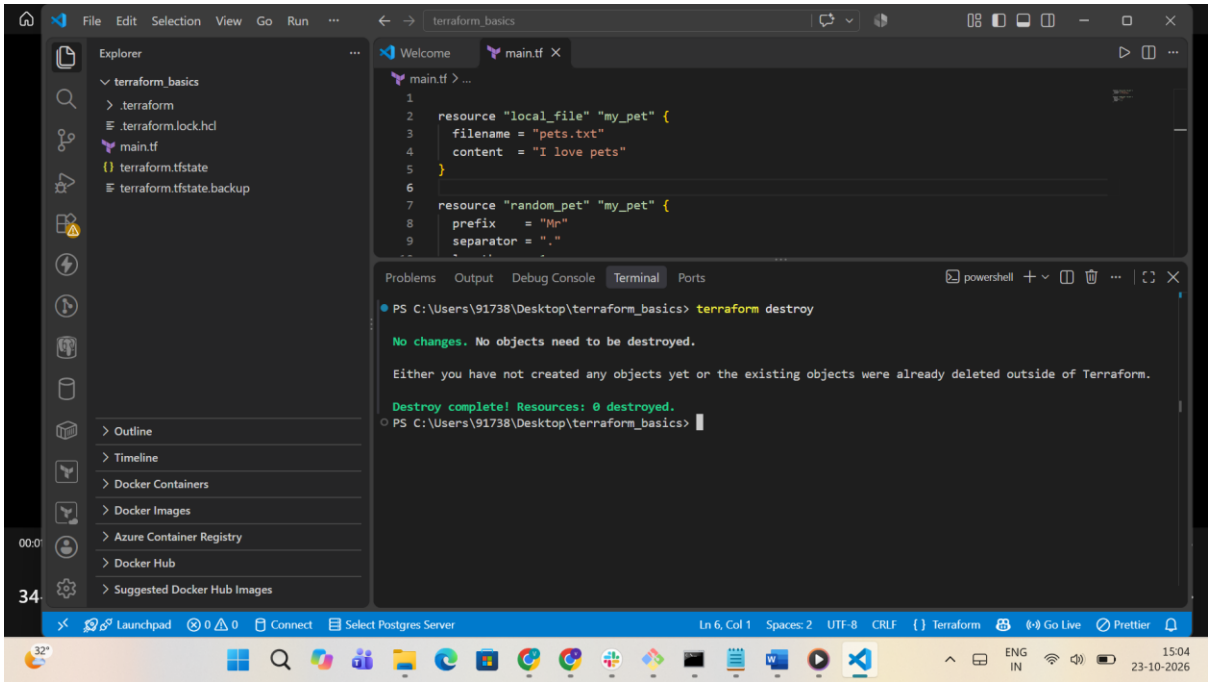
Tasks on Terraform-02

Question 1: Watch the Terraform-02 Video

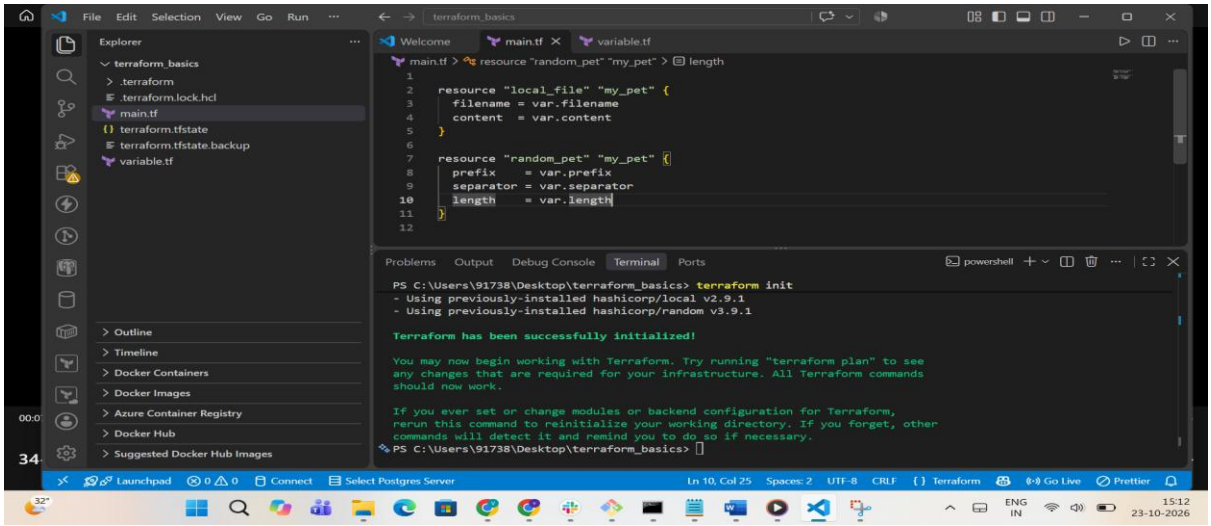
Watching video of terraform-02 completed.

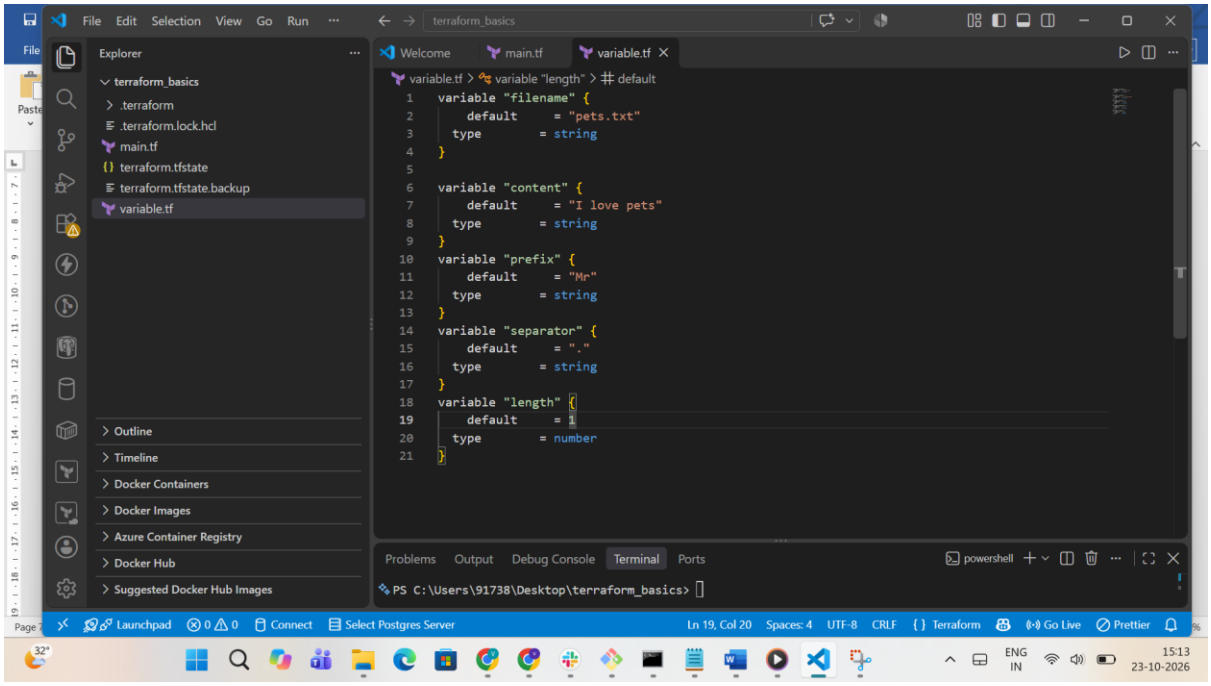
Question 2: Execute All the Templates Shown in the Video

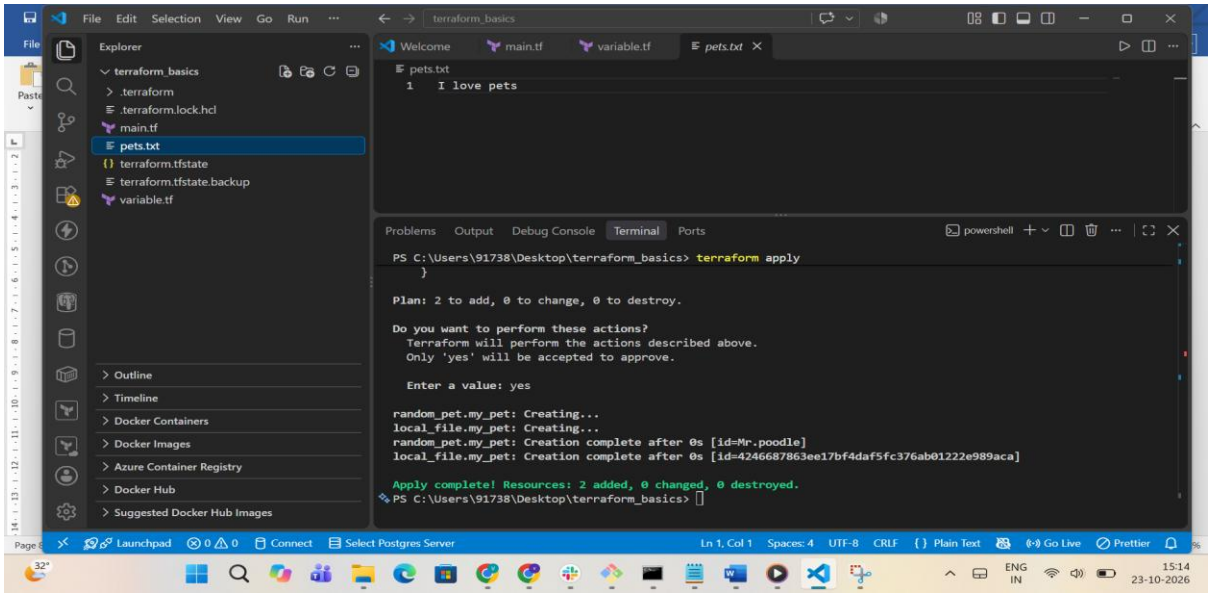
1. Terraform destroy

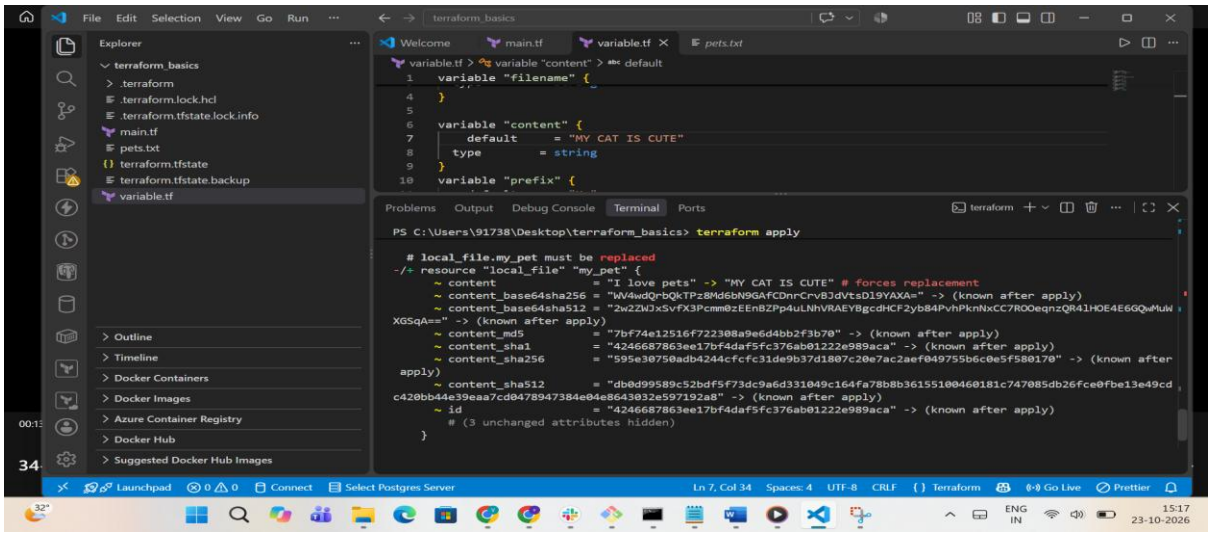


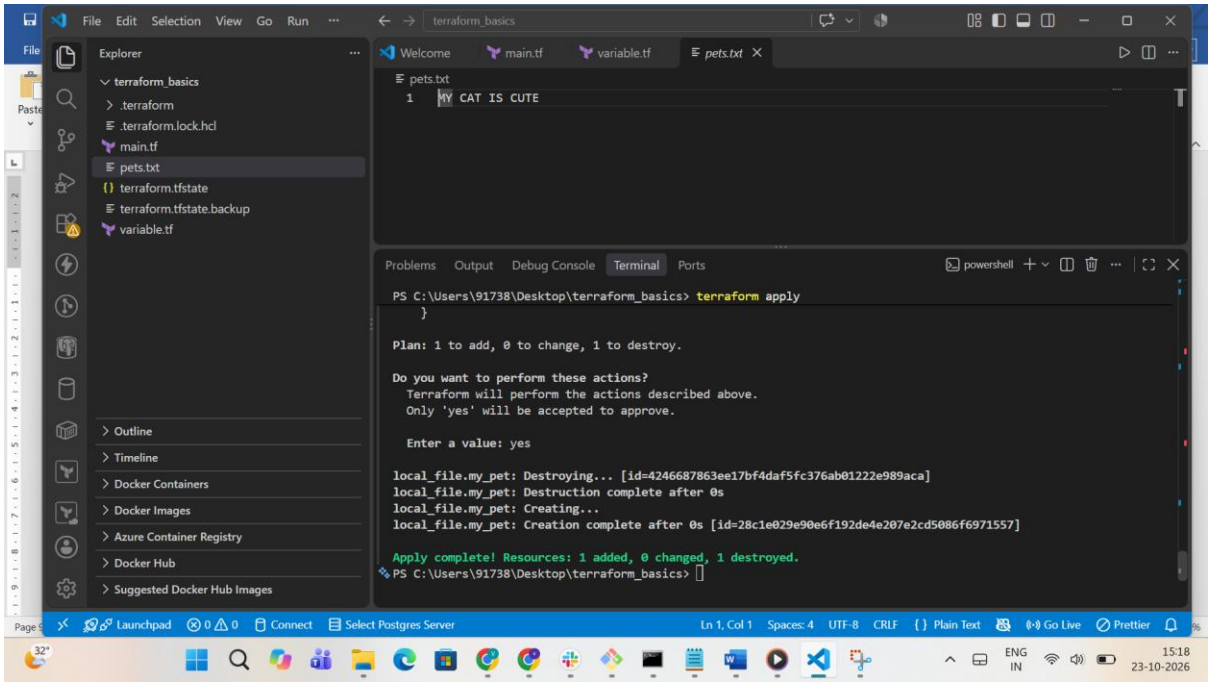
2. variable.tf file




File Edit Selection View Go Run
terraform_basics
Explorer
terraform_basics
.terraform
.terraform.lock.hcl
main.tf
terraform.tfstate
terraform.tfstate.backup
variable.tf
Welcome
main.tf
variable.tf
variable.tf > variable "length" > default
variable "filename" {
default = "pets.txt"
type = string
}
variable "content" {
default = "I love pets"
type = string
}
variable "prefix" {
default = "Mr"
type = string
}
variable "separator" {
default = "."
type = string
}
variable "length" {
default = 1
type = number
}
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform_basics>
Launchpad
Connect
Select Postgres Server
Ln 19, Col 20 Spaces: 4 UTF-8 CRLF { } Terraform
Go Live
Prettier
ENG IN
15:13
23-10-2026

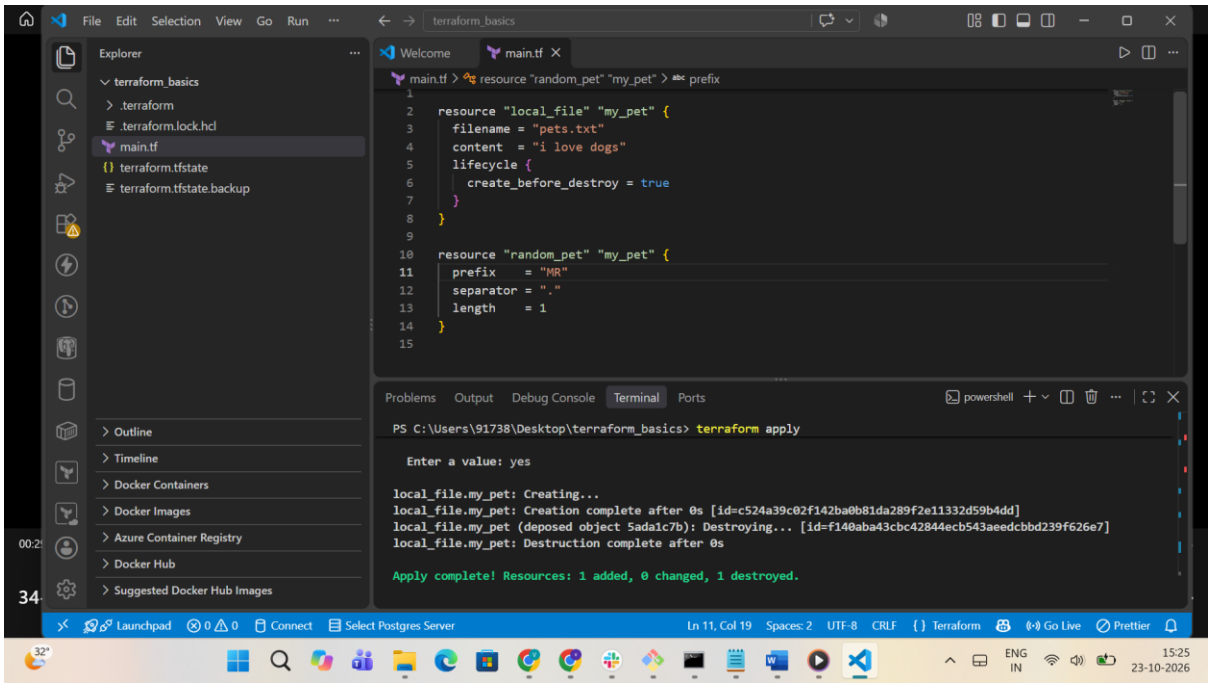

File Edit Selection View Go Run
terraform_basics
Explorer
terraform_basics
.terraform
.terraform.lock.hcl
main.tf
pets.txt
terraform.tfstate
terraform.tfstate.backup
variable.tf
Welcome
main.tf
variable.tf
pets.txt
pets.txt
1 I love pets
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Problems Output Debug Console Terminal Ports
powershell
PS C:\Users\91738\Desktop\terraform_basics> terraform apply
}
Plan: 2 to add, 0 to change, 0 to destroy.
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
random_pet.my_pet: Creating...
local_file.my_pet: Creating...
random_pet.my_pet: Creation complete after 0s [id=Mr.poodle]
local_file.my_pet: Creation complete after 0s [id=4246687863ee17bf4daf5fc376ab01222e989aca]
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS C:\Users\91738\Desktop\terraform_basics>
Launchpad
Connect
Select Postgres Server
Ln 1, Col 1 Spaces: 4 UTF-8 CRLF { } Plain Text
Go Live
Prettier
ENG IN
15:14
23-10-2026

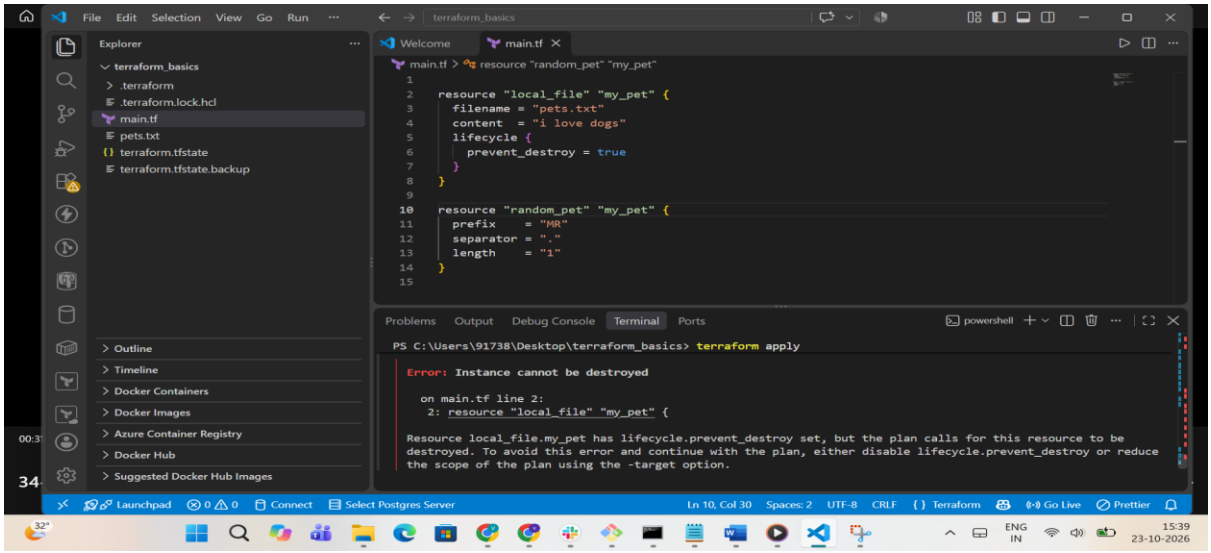

File Edit Selection View Go Run
terraform_basics
Explorer
terraform_basics
.terraform
.terraform.lock.hcl
.terraform.tfstate.lock.info
main.tf
pets.txt
terraform.tfstate
terraform.tfstate.backup
variable.tf
Welcome
main.tf
variable.tf
pets.txt
variable.tf > variable "content" > default
variable "filename" {
}
variable "content" {
default = "MY CAT IS CUTE"
type = string
}
variable "prefix" {
Outline
Timeline
Docker Containers
Docker Images
Azure Container Registry
Docker Hub
Suggested Docker Hub Images
Problems Output Debug Console Terminal Ports
terraform
PS C:\Users\91738\Desktop\terraform_basics> terraform apply
local_file.my_pet must be replaced
-/+ resource "local_file" "my_pet" {
~ content = "I love pets" -> "MY CAT IS CUTE" # forces replacement
~ content_base64sha256 = "WV4wdQrbQkTPz8Md6bN9GAfCDnrCrvBJdVtsD19YAXA=" -> (known after apply)
~ content_base64sha512 = "2w2ZWJxSvfX3Pcmm0zEEnBZPp4uLNhVRAEYBgcdHCF2yb84PvhPknNxCC7ROOeqnzQR41HOE4E6GQwMuWiXGSqA==" -> (known after apply)
~ content_md5 = "7bf74e12516f722308a9e6d4bb2f3b70" -> (known after apply)
~ content_sha1 = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
~ content_sha256 = "595e30750adb4244cfcfc31de9b37d1807c20e7ac2aef049755b6c0e5f580170" -> (known after apply)
~ content_sha512 = "db0d99589c52bdf5f73dc9a6d331049c164fa78b8b36155100460181c747085db26fce0fbe13e49cdc420bb44e39eaa7cd0478947384e04e8643032e597192a8" -> (known after apply)
~ id = "4246687863ee17bf4daf5fc376ab01222e989aca" -> (known after apply)
(3 unchanged attributes hidden)
}
Launchpad
Connect
Select Postgres Server
Ln 7, Col 34 Spaces: 4 UTF-8 CRLF { } Terraform
Go Live
Prettier
ENG IN
15:17
23-10-2026

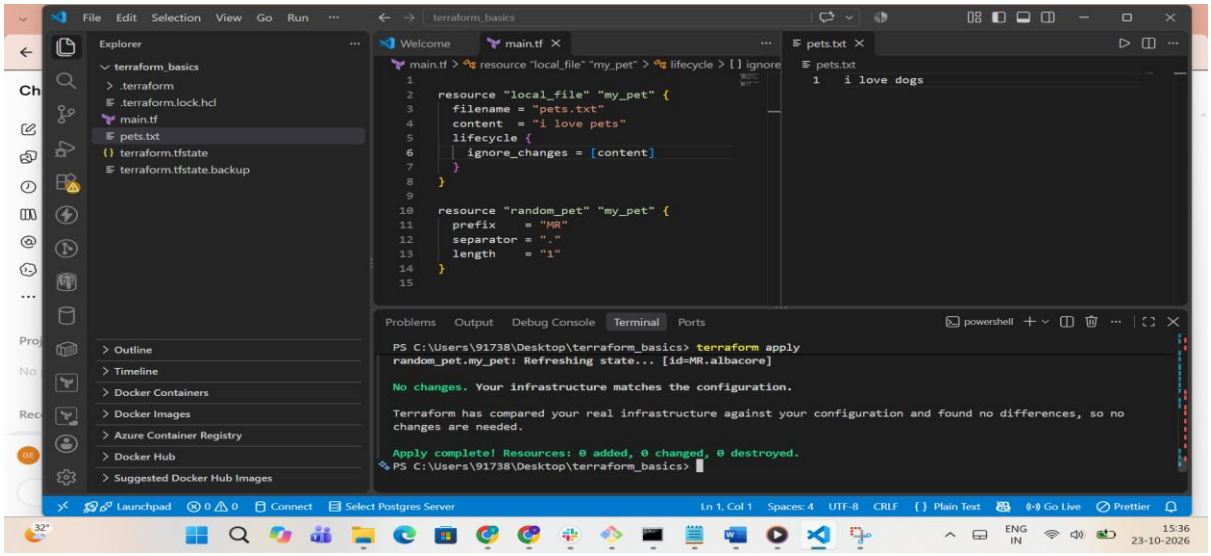


3.Life cycle rule:

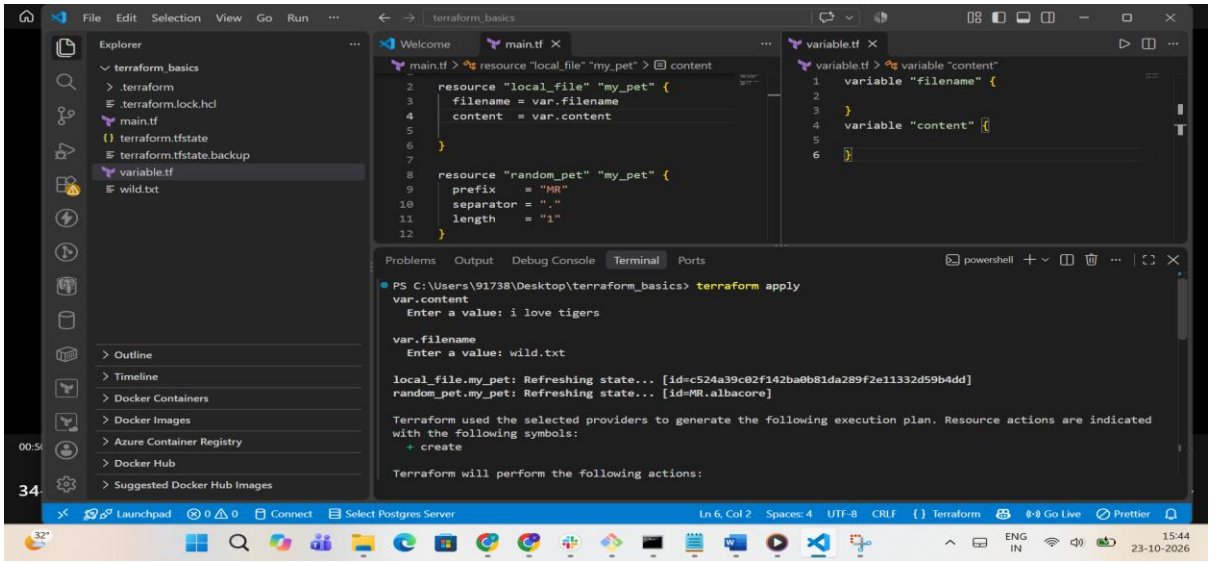
create_before_destroy, prevent_destroy, ignore_changes

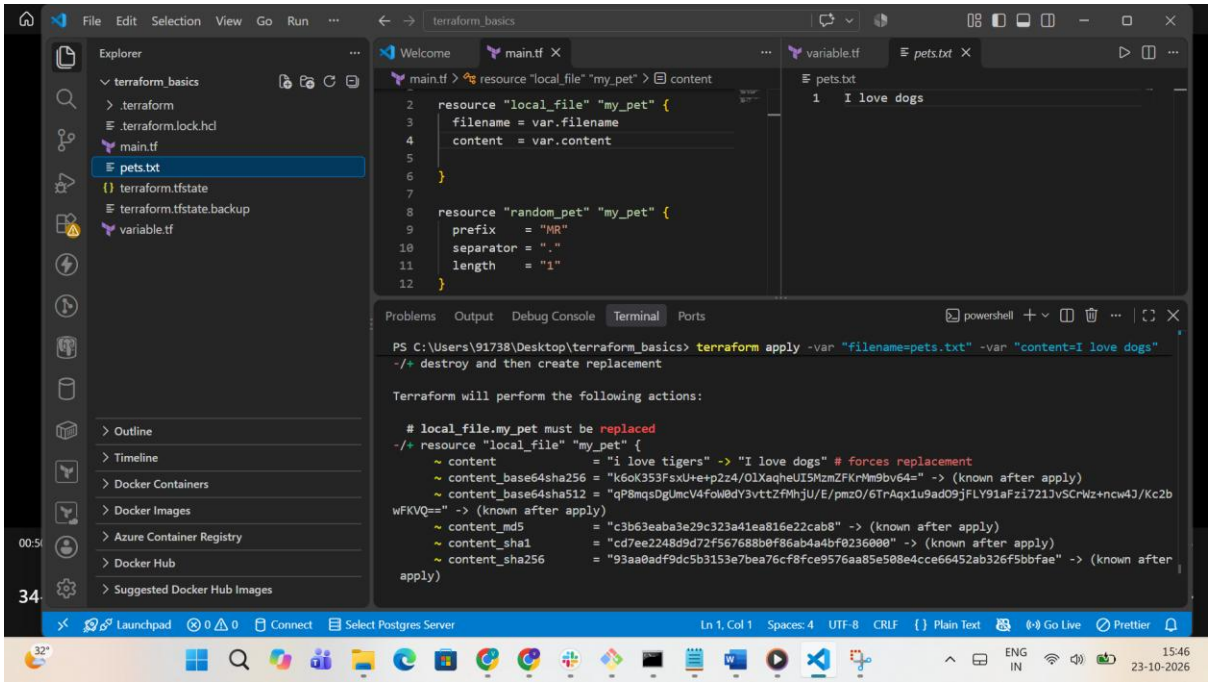






4. Different ways to use variables





Question 3: Integrate Terraform into Jenkins Using the Terraform Plugin

Implementations steps:

1. Launch an EC2 instance using Amazon Linux.
2. Connect to the EC2 instance through SSH.
3. Install Java and Jenkins.
4. Start Jenkins and enable it to run automatically:

   sudo systemctl enable jenkins

   sudo systemctl start jenkins
5. Configure the EC2 security group to allow access to the Jenkins web interface.
6. Open Jenkins in a browser using the EC2 public IP and configured port.
7. Complete the initial Jenkins setup and log in as the administrator.
8. Navigate to Manage Jenkins → Plugins.
9. Install the Terraform plugin.
10. Install Terraform on the EC2 instance and verify the installation:

    /usr/bin/terraform
11. Navigate to Manage Jenkins → Tools and configure the Terraform installation using the Terraform executable path:

    terraform {

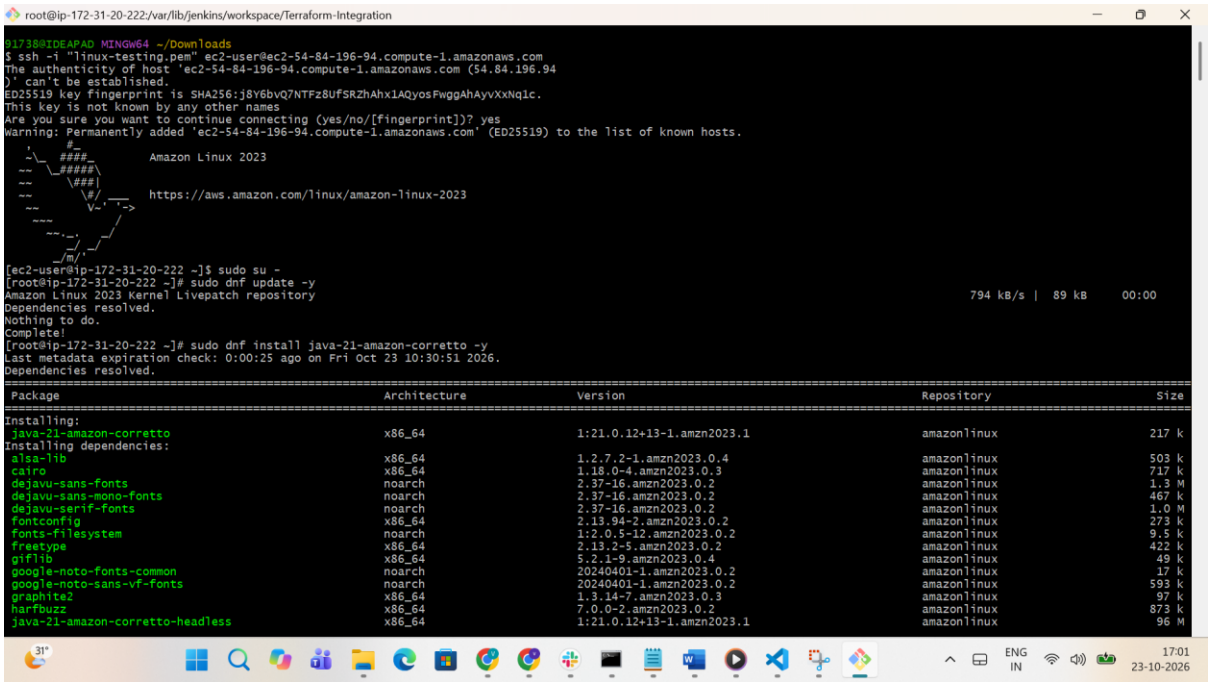
```
  required_providers {
    local = {
      source  = "hashicorp/local"
      version = "~> 2.5"
    }
  }
}

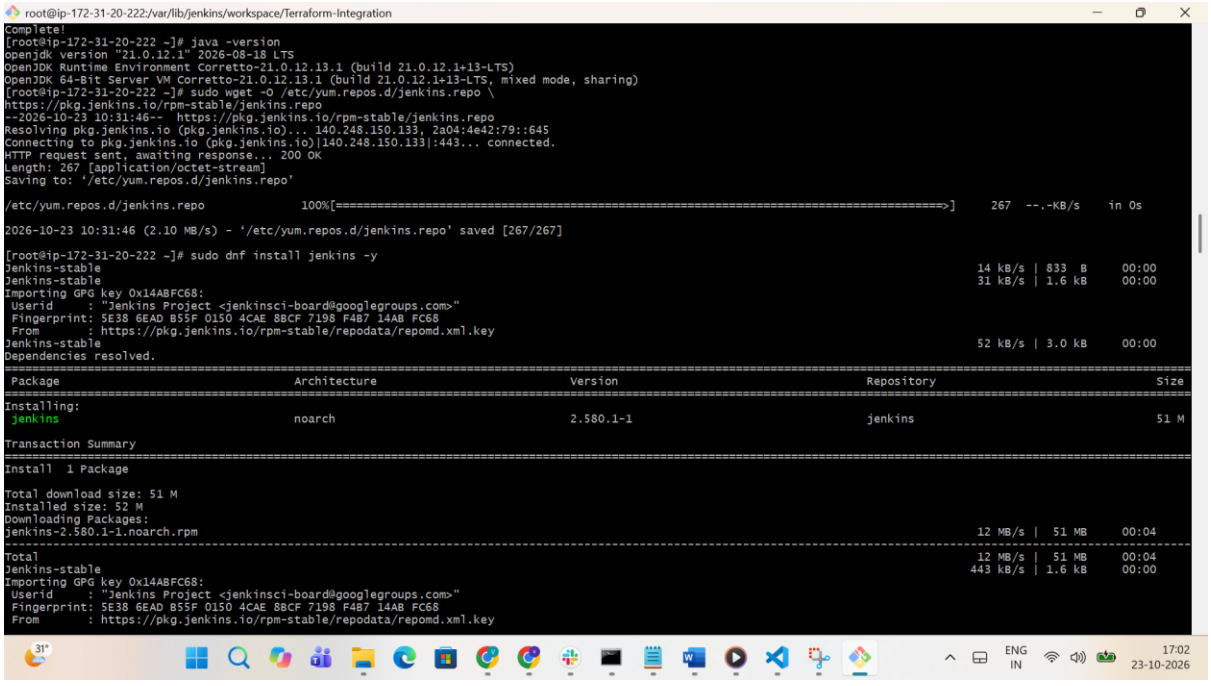
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "i love dogs"
}
```

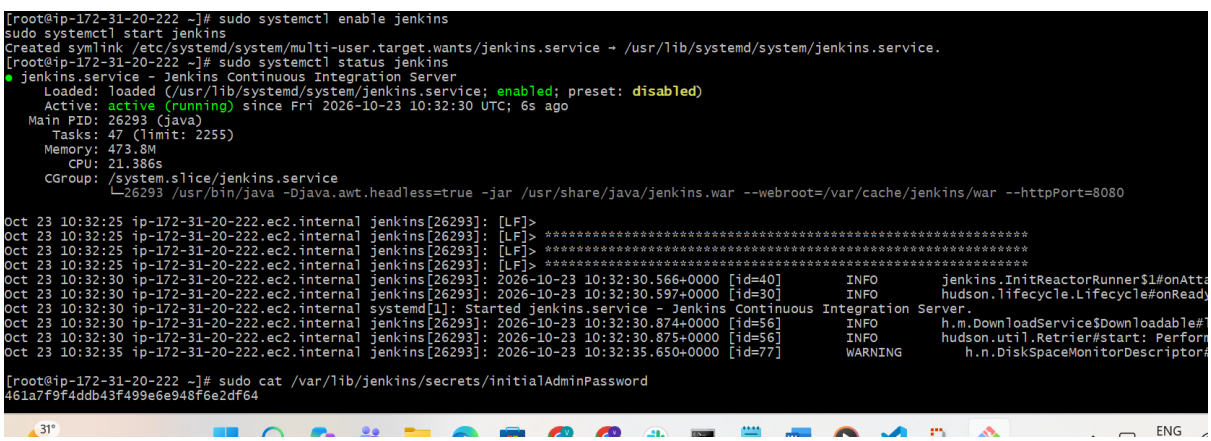
12. Create a Freestyle project named Terraform-Integration.
13. Open the job configuration and add the Terraform build step.
14. Select the configured Terraform installation.
15. Create a main.tf file in the job workspace:

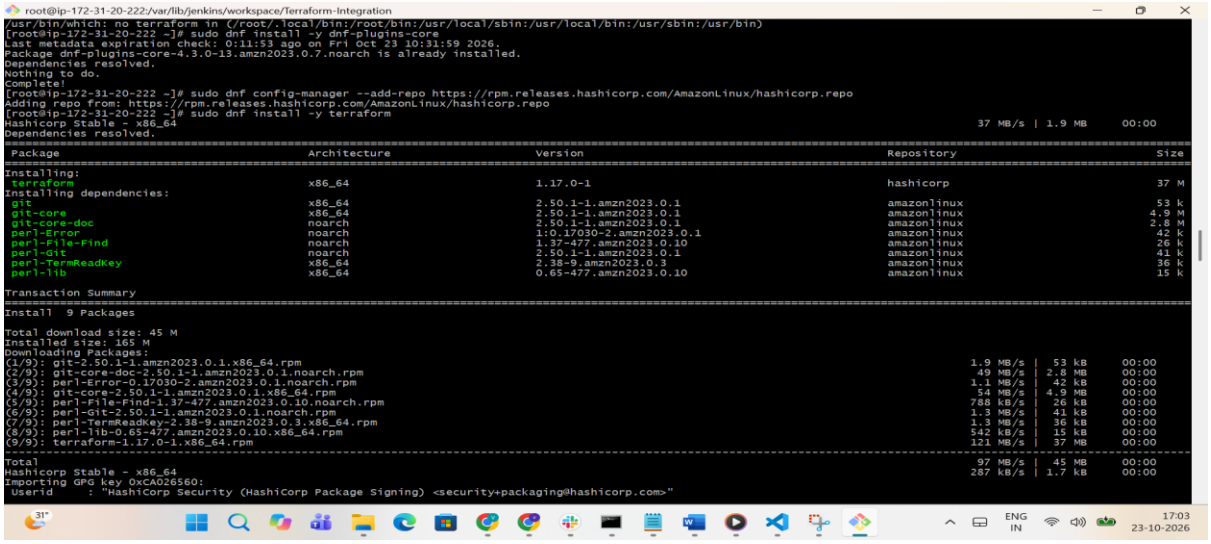
    sudo -u jenkins /usr/bin/terraform init
16. Set the Terraform Configuration Path to ..
17. Ensure the Jenkins user has permission to access the job workspace and Terraform files.
18. Initialize the Terraform configuration and download the required provider:

    cat /var/lib/jenkins/workspace/Terraform-Integration/pets.txt
19. Configure the Jenkins job to execute Terraform commands through the Terraform plugin.
20. Configure a non-interactive apply command using terraform apply -auto-approve where supported by the job configuration.
21. Execute the Jenkins job using Build Now.
22. Open Console Output and verify the execution results.
23. Resolve the Terraform state-file permission error by correcting the ownership of the workspace files, where necessary.
24. Verify the created file:
25. Confirm that the file contains i love dogs and that the Jenkins build completes successfully.


root@ip-172-31-20-222:/var/lib/jenkins/workspace/Terraform-Integration
9173@IDEAPAD MINGW64 ~/Downloads
$ ssh -i "linux-testing.pem" ec2-user@ec2-54-84-196-94.compute-1.amazonaws.com
The authenticity of host 'ec2-54-84-196-94.compute-1.amazonaws.com (54.84.196.94
)' can't be established.
ED25519 key fingerprint is SHA256:j8Y6bvQ7NTFz8UfSRZhAhx1AQyosFwggAhAyvXxNq1c.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-84-196-94.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-172-31-20-222 ~]$ sudo su -
[root@ip-172-31-20-222 ~]# sudo dnf update -y
Amazon Linux 2023 Kernel Livepatch repository 794 kB/s | 89 kB 00:00
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-20-222 ~]# sudo dnf install java-21-amazon-corretto -y
Last metadata expiration check: 0:00:25 ago on Fri Oct 23 10:30:51 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
java-21-amazon-corretto x86_64 1:21.0.12+13-1.amzn2023.1 amazonlinux 217 k
Installing dependencies:
alsa-lib x86_64 1.2.7.2-1.amzn2023.0.4 amazonlinux 503 k
cairo x86_64 1.18.0-4.amzn2023.0.3 amazonlinux 717 k
dejavu-sans-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.3 M
dejavu-sans-mono-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 467 k
dejavu-serif-fonts noarch 2.37-16.amzn2023.0.2 amazonlinux 1.0 M
fontconfig x86_64 2.13.94-2.amzn2023.0.2 amazonlinux 273 k
fonts-filesystem noarch 1:2.0.5-12.amzn2023.0.2 amazonlinux 9.5 k
freetype x86_64 2.13.2-5.amzn2023.0.2 amazonlinux 422 k
giflib x86_64 5.2.1-9.amzn2023.0.4 amazonlinux 49 k
google-noto-fonts-common noarch 20240401-1.amzn2023.0.2 amazonlinux 17 k
google-noto-sans-vf-fonts noarch 20240401-1.amzn2023.0.2 amazonlinux 593 k
graphite2 x86_64 1.3.14-7.amzn2023.0.3 amazonlinux 97 k
harfbuzz x86_64 7.0.0-2.amzn2023.0.2 amazonlinux 873 k
java-21-amazon-corretto-headless x86_64 1:21.0.12+13-1.amzn2023.1 amazonlinux 96 M
ENG IN 17:01 23-10-2026


root@ip-172-31-20-222:/var/lib/jenkins/workspace/Terraform-Integration
Complete!
[root@ip-172-31-20-222 ~]# java -version
openjdk version "21.0.12.1" 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-21.0.12.13.1 (build 21.0.12.1+13-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.13.1 (build 21.0.12.1+13-LTS, mixed mode, sharing)
[root@ip-172-31-20-222 ~]# sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/rpm-stable/jenkins.repo
--2026-10-23 10:31:46-- https://pkg.jenkins.io/rpm-stable/jenkins.repo
Resolving pkg.jenkins.io (pkg.jenkins.io)... 140.248.150.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|140.248.150.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 267 [application/octet-stream]
Saving to: '/etc/yum.repos.d/jenkins.repo'
/etc/yum.repos.d/jenkins.repo 100%[=>] 267 --.-KB/s in 0s
2026-10-23 10:31:46 (2.10 MB/s) - '/etc/yum.repos.d/jenkins.repo' saved [267/267]
[root@ip-172-31-20-222 ~]# sudo dnf install jenkins -y
Jenkins-stable 14 kB/s | 833 B 00:00
Jenkins-stable 31 kB/s | 1.6 kB 00:00
Importing GPG key 0x14ABFC68:
Userid : "Jenkins Project <jenkinsci-board@googlegroups.com>"
Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
From : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
Jenkins-stable 52 kB/s | 3.0 kB 00:00
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
jenkins noarch 2.580.1-1 jenkins 51 M
Transaction Summary
Install 1 Package
Total download size: 51 M
Installed size: 52 M
Downloading Packages:
jenkins-2.580.1-1.noarch.rpm 12 MB/s | 51 MB 00:04
Total 12 MB/s | 51 MB 00:04
Jenkins-stable 443 kB/s | 1.6 kB 00:00
Importing GPG key 0x14ABFC68:
Userid : "Jenkins Project <jenkinsci-board@googlegroups.com>"
Fingerprint: 5E38 6EAD B55F 0150 4CAE 8BCF 7198 F4B7 14AB FC68
From : https://pkg.jenkins.io/rpm-stable/repodata/repomd.xml.key
ENG IN 17:02 23-10-2026


[root@ip-172-31-20-222 ~]# sudo systemctl enable jenkins
sudo systemctl start jenkins
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
[root@ip-172-31-20-222 ~]# sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: disabled)
Active: active (running) since Fri 2026-10-23 10:32:30 UTC; 6s ago
Main PID: 26293 (java)
Tasks: 47 (limit: 2255)
Memory: 473.8M
CPU: 21.386s
CGroup: /system.slice/jenkins.service
└─26293 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
Oct 23 10:32:25 ip-172-31-20-222.ec2.internal jenkins[26293]: [LF]>
Oct 23 10:32:25 ip-172-31-20-222.ec2.internal jenkins[26293]: [LF]> **
Oct 23 10:32:25 ip-172-31-20-222.ec2.internal jenkins[26293]: [LF]> **
Oct 23 10:32:25 ip-172-31-20-222.ec2.internal jenkins[26293]: [LF]> **
Oct 23 10:32:30 ip-172-31-20-222.ec2.internal jenkins[26293]: 2026-10-23 10:32:30.566+0000 [id=40] INFO jenkins.InitReactorRunner$1#onAtta
Oct 23 10:32:30 ip-172-31-20-222.ec2.internal jenkins[26293]: 2026-10-23 10:32:30.597+0000 [id=30] INFO hudson.lifecycle.Lifecycle#onReady
Oct 23 10:32:30 ip-172-31-20-222.ec2.internal systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Oct 23 10:32:30 ip-172-31-20-222.ec2.internal jenkins[26293]: 2026-10-23 10:32:30.874+0000 [id=56] INFO h.m.DownloadService$Downloadable#
Oct 23 10:32:30 ip-172-31-20-222.ec2.internal jenkins[26293]: 2026-10-23 10:32:30.875+0000 [id=56] INFO hudson.util.Retrier#start: Perfor
Oct 23 10:32:35 ip-172-31-20-222.ec2.internal jenkins[26293]: 2026-10-23 10:32:35.650+0000 [id=77] WARNING h.n.DiskSpaceMonitorDescriptor#
[root@ip-172-31-20-222 ~]# sudo cat /var/lib/jenkins/secrets/initialAdminPassword
461a7f9f4ddb43f499e6e948f6e2df64
ENG

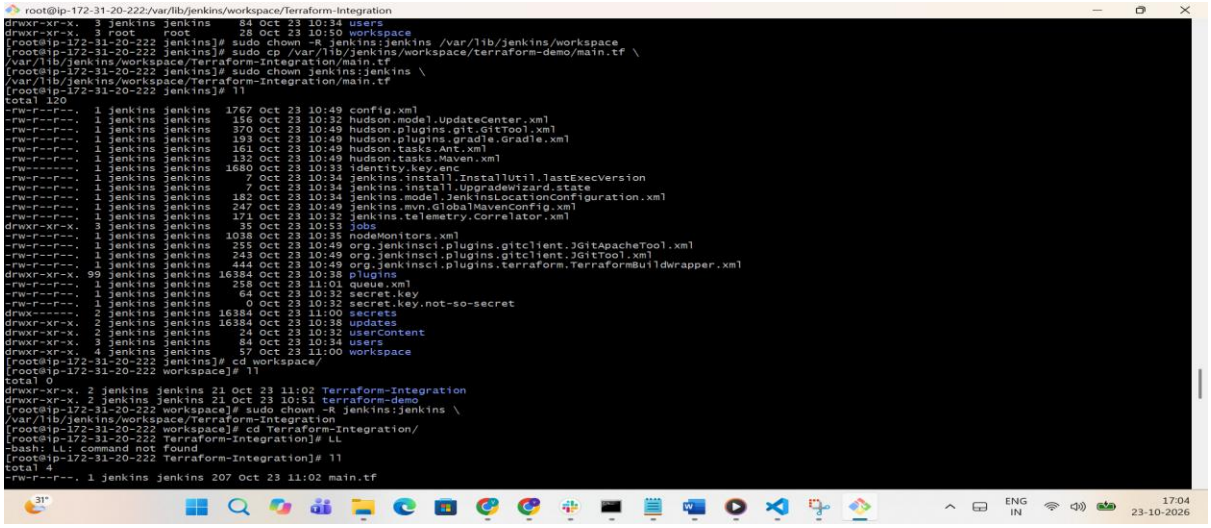



```
Installed:
  git-2.50.1-1.amzn2023.0.1.x86_64            git-core-2.50.1-1.amzn2023.0.1.x86_64 git-core-doc-2.50.1-1.amzn2023.0.1.noarch   perl-Error-1:0.17030-2.amzn2023.0.1.noa
  perl-File-Find-1.37-477.amzn2023.0.10.noarch perl-Git-2.50.1-1.amzn2023.0.1.noarch perl-TermReadKey-2.38-9.amzn2023.0.3.x86_64 perl-lib-0.65-477.amzn2023.0.10.x86_64
  terraform-1.17.0-1.x86_64

Complete!
[root@ip-172-31-20-222 ~]# terraform version
Terraform v1.17.0
on linux_amd64

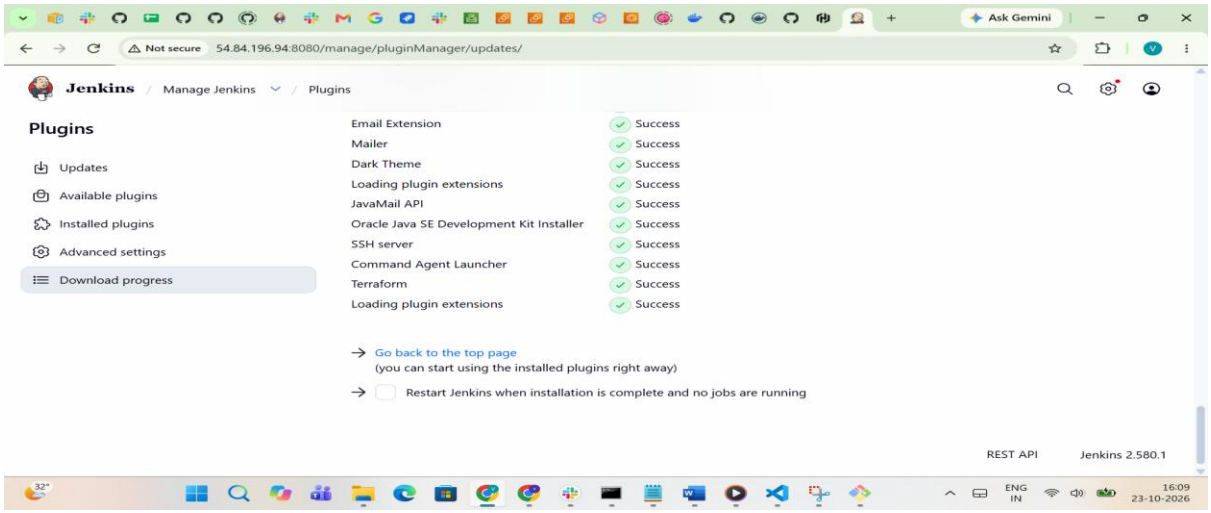
[root@ip-172-31-20-222 ~]# which terraform
/usr/bin/terraform
[root@ip-172-31-20-222 ~]# sudo -u jenkins terraform version
Terraform v1.17.0
on linux_amd64

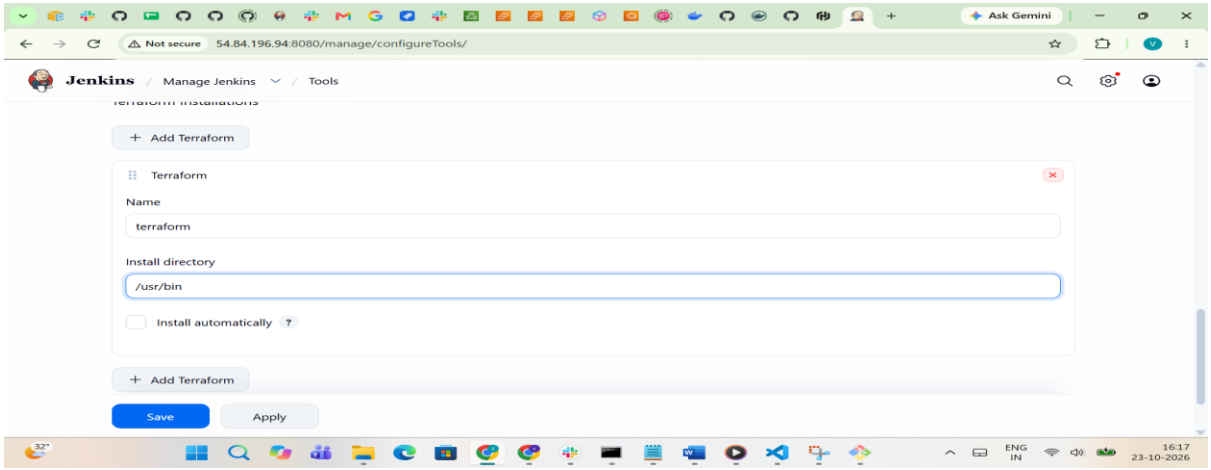
[root@ip-172-31-20-222 ~]# readlink -f "$(which terraform)"
/usr/bin/terraform
[root@ip-172-31-20-222 ~]# dirname "$(readlink -f "$(which terraform)")"
/usr/bin
[root@ip-172-31-20-222 ~]# ls -l /opt/terraform
ls: cannot access '/opt/terraform': No such file or directory
[root@ip-172-31-20-222 ~]# ^C
[root@ip-172-31-20-222 ~]# cd /usr
[root@ip-172-31-20-222 usr]# ll
total 176
dr-xr-xr-x.  2 root root 32768 Oct 23 10:44 bin
drwxr-xr-x.  2 root root     6 Jan 30  2023 games
drwxr-xr-x.  3 root root    23 Oct 21 17:47 include
dr-xr-xr-x. 32 root root 16384 Oct 23 10:31 lib
dr-xr-xr-x. 44 root root 32768 Oct 23 10:31 lib64
drwxr-xr-x. 27 root root 16384 Oct 23 10:44 libexec
drwxr-xr-x. 12 root root   131 Oct 21 17:47 local
dr-xr-xr-x.  2 root root 16384 Oct 21 17:49 sbin
drwxr-xr-x. 98 root root 16384 Oct 23 10:44 share
drwxr-xr-x.  4 root root    34 Oct 21 17:47 src
lrwxrwxrwx.  1 root root    10 Jan 30  2023 tmp -> ../var/tmp
[root@ip-172-31-20-222 usr]# ^C
[root@ip-172-31-20-222 usr]# sudo -u jenkins terraform version
Terraform v1.17.0
on linux_amd64
```

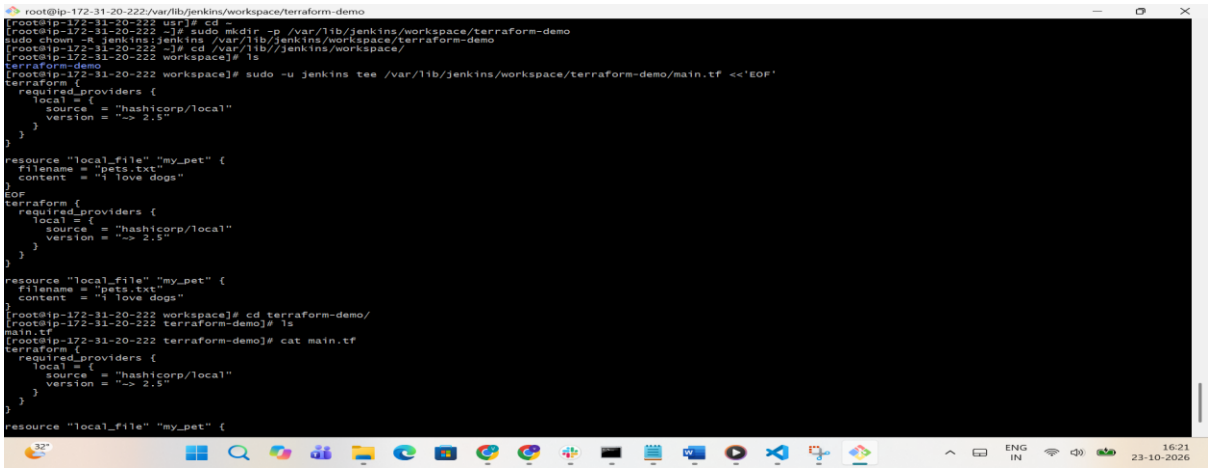


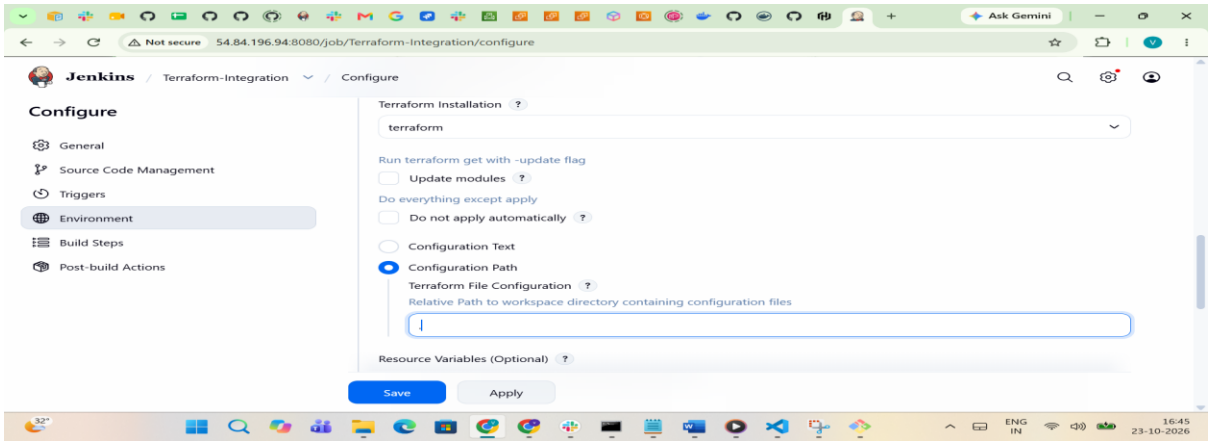

```
[root@ip-172-31-20-222 Terraform-Integration]# cat main.tf
terraform {
  required_providers {
    local = {
      source  = "hashicorp/local"
      version = "~> 2.5"
    }
  }
}

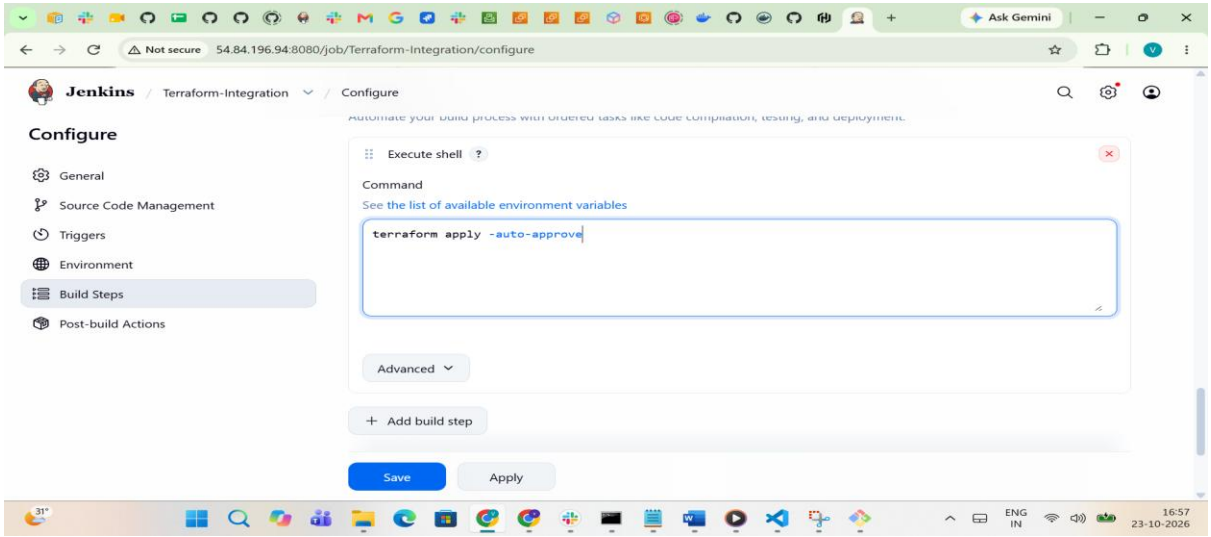
resource "local_file" "my_pet" {
  filename = "pets.txt"
  content  = "i love dogs"
}
[root@ip-172-31-20-222 Terraform-Integration]#
```

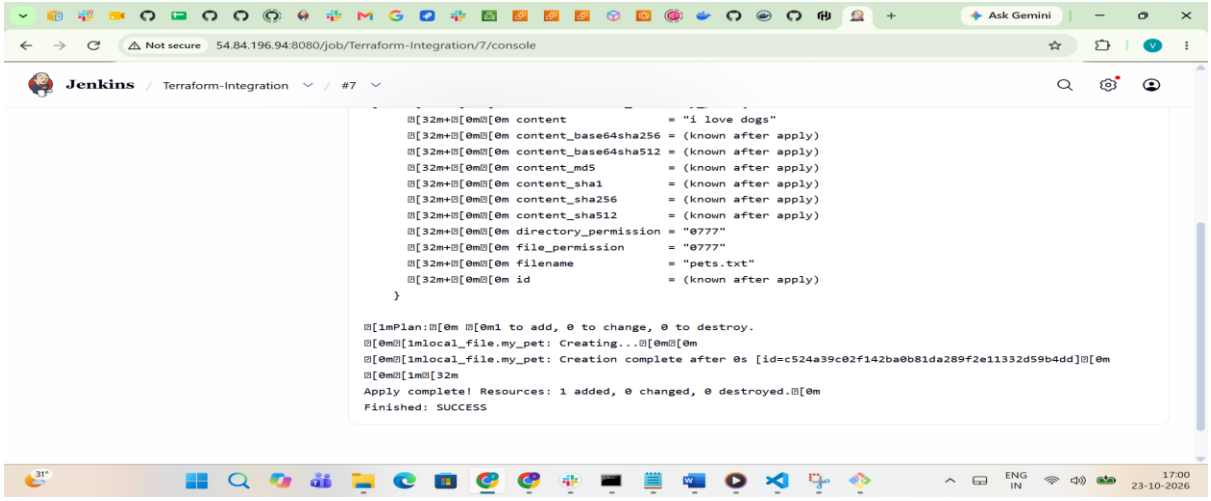

Ask Gemini
Not secure 54.84.196.94:8080/manage/pluginManager/updates/
Jenkins / Manage Jenkins / Plugins
Plugins
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
Email Extension Success
Mailer Success
Dark Theme Success
Loading plugin extensions Success
JavaMail API Success
Oracle Java SE Development Kit Installer Success
SSH server Success
Command Agent Launcher Success
Terraform Success
Loading plugin extensions Success
Go back to the top page
(you can start using the installed plugins right away)
Restart Jenkins when installation is complete and no jobs are running
REST API
Jenkins 2.580.1


Ask Gemini
Not secure 54.84.196.94:8080/manage/configureTools/
Jenkins / Manage Jenkins / Tools
Add Terraform
Terraform
Name
terraform
Install directory
/usr/bin
Install automatically
Add Terraform
Save
Apply


root@ip-172-31-20-222:/var/lib/jenkins/workspace/terraform-demo
[root@ip-172-31-20-222 usr]# cd ~
[root@ip-172-31-20-222 ~]# sudo mkdir -p /var/lib/jenkins/workspace/terraform-demo
sudo chown -R jenkins:jenkins /var/lib/jenkins/workspace/terraform-demo
[root@ip-172-31-20-222 ~]# cd /var/lib/jenkins/workspace/
[root@ip-172-31-20-222 workspace]# ls
terraform-demo
[root@ip-172-31-20-222 workspace]# sudo -u jenkins tee /var/lib/jenkins/workspace/terraform-demo/main.tf <<'EOF'
terraform {
required_providers {
local = {
source = "hashicorp/local"
version = "~> 2.5"
}
}
}
resource "local_file" "my_pet" {
filename = "pets.txt"
content = "I love dogs"
}
EOF
terraform {
required_providers {
local = {
source = "hashicorp/local"
version = "~> 2.5"
}
}
}
resource "local_file" "my_pet" {
filename = "pets.txt"
content = "I love dogs"
}
[root@ip-172-31-20-222 workspace]# cd terraform-demo/
[root@ip-172-31-20-222 terraform-demo]# ls
main.tf
[root@ip-172-31-20-222 terraform-demo]# cat main.tf
terraform {
required_providers {
local = {
source = "hashicorp/local"
version = "~> 2.5"
}
}
}
resource "local_file" "my_pet" {








```
[root@ip-172-31-20-222 Terraform-Integration]# ll
total 12
-rw-r--r--. 1 jenkins jenkins  207 Oct 23 11:02 main.tf
-rwxr-xr-x. 1 jenkins jenkins   11 Oct 23 11:30 pets.txt
drwxr-xr-x. 3 jenkins jenkins   24 Oct 23 11:30 terraform-plugin
-rw-r--r--. 1 jenkins jenkins 1622 Oct 23 11:30 terraform.tfstate
[root@ip-172-31-20-222 Terraform-Integration]# cat pets.txt
i love dogs[root@ip-172-31-20-222 Terraform-Integration]# ^C
```